

39.3 Bayesian thread IV: what carries over

The Bayesian thread has three instalments behind it. [Section 7.6](#) put a normal-inverse-gamma prior on $(\boldsymbol{\beta}, \sigma^2)$ in the normal linear model and found the posterior in closed form (Theorem 7.6.2): precisions add, the posterior mean is a precision-weighted average of the prior mean and the least squares estimate (Proposition 7.6.3), and the improper prior $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ reproduces least squares and its t intervals exactly (Corollary 7.6.4). [Section 12.6](#) summarized that posterior by credible sets (Definition 12.6.1), and [Section 32.6](#) read a random effect as a prior, showed that the BLUP is a posterior mean and REML a marginal posterior mode (Proposition 32.6.1), and ran a Gibbs sampler to do what the closed form could not. This short section says which of those survive the move to a generalized linear model, and why the others do not.

39.3.1 What survives

Three things carry over unchanged, being definitions rather than computations. **The posterior is still prior times likelihood:** Bayes' theorem does not know what family $\mathbf{y}$ came from. **Credible sets still mean what they meant:** Definition 12.6.1 is a statement about a posterior distribution, not about a normal one. **A random effect is still a prior:** the generalized linear mixed models of [Chapter 40](#) are Eq. 32.6.1 with a nonnormal first level, and the empirical Bayes reading (Proposition 27.5.4) survives with them.

39.3.2 What does not, and why

Everything that made [Section 7.6](#) *closed form* rests on one fact: the normal log-likelihood is a quadratic in $\boldsymbol{\beta}$, so prior times likelihood is again the exponential of a quadratic and completing the square finishes the job. Nothing weaker will do.

Proposition 39.3.1 — Why the conjugate machinery stops at the normal family

Let $\mathbf{y}$ follow the exponential dispersion family of Definition 34.2.1 with the canonical link, so that $\theta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ and $\ell(\boldsymbol{\beta}) = \phi^{-1} \sum_i w_i \{y_i \theta_i - b(\theta_i)\} + c$. Suppose b is twice continuously differentiable on an open interval Θ . Then the following are equivalent.

- (a) For every n , every $\mathbf{X}$ with $\text{rank}(\mathbf{X}) = p$ and every $\mathbf{y}$, the function $\boldsymbol{\beta} \mapsto \ell(\boldsymbol{\beta})$ is a quadratic polynomial on $\{\boldsymbol{\beta} : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} \in \Theta\}$.
- (b) b'' is constant on Θ .
- (c) The variance function is constant: $V(\mu) = b''(\theta)$ does not depend on μ , so the family is normal (Exercise (B2)).

Consequently, for every family except the normal, a normal prior on $\boldsymbol{\beta}$ gives a posterior that is not normal and has no closed-form normalizing constant.

Proof. The Hessian of ℓ is $-\phi^{-1} \mathbf{X}^\top \text{diag}\{w_i b''(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\} \mathbf{X}$ by Theorem 34.3.1(d). If (b) holds this is constant in $\boldsymbol{\beta}$, and a function with a constant Hessian is a quadratic polynomial, giving (a). Conversely, if (a) holds, take $p = 1$, $n = 1$, $w_1 = 1$, $x_{11} = 1$:

the Hessian is the scalar $-b''(\beta)/\phi$, which must then be constant over Θ , which is (b). The equivalence of (b) and (c) is the definition $V(\mu) = b''(\theta)$ of Definition 34.1.2 with $\mu = b'(\theta)$, which makes $\theta \mapsto \mu$ a bijection: $b'' \equiv \sigma^2$ gives $b(\theta) = \sigma^2 \theta^2/2$ up to an affine term, the normal cumulant function.

For the last sentence, with π the $N_p(\mathbf{m}, \Sigma)$ density the log posterior is $\ell(\boldsymbol{\beta}) - \frac{1}{2}(\boldsymbol{\beta} - \mathbf{m})^\top \Sigma^{-1}(\boldsymbol{\beta} - \mathbf{m})$ up to a constant; were it the log of a normal density it would be a quadratic in $\boldsymbol{\beta}$, hence so would ℓ , which forces the normal family.

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So the failure is not a want of cleverness. It is the same failure that made the likelihood equations of Theorem 34.3.1 nonlinear and forced iteratively reweighted least squares on the frequentist side: the score is not linear in $\boldsymbol{\beta}$, so the posterior is not normal, and the integrals defining the posterior mean, the posterior variance and the marginal likelihood

$$m(\mathbf{y}) = \int L(\boldsymbol{\beta}; \mathbf{y}) \pi(\boldsymbol{\beta}) d\boldsymbol{\beta} \quad (39.3.1)$$

have no closed form.

39.3.3 Conjugate priors do exist

It is worth being precise, because “generalized linear models have no conjugate priors” is false as stated. Diaconis and Ylvisaker (1979) characterized the conjugate priors of an exponential family; the canonical-link likelihood is an exponential family in $\boldsymbol{\beta}$, and their construction gives the density

$$\pi(\boldsymbol{\beta} \mid \mathbf{m}_0, \kappa_0) \propto \exp \left[\kappa_0^{-1} \{ \mathbf{m}_0^\top \mathbf{X} \boldsymbol{\beta} - \sum_i v_i b(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}) \} \right] \quad (39.3.2)$$

in the form developed for generalized linear models by Bedrick, Christensen and Johnson (1996) and Chen and Ibrahim (2003), with a pseudo-response vector $\mathbf{m}_0$ and a prior weight v_i on each observation. It has the shape of the likelihood, so the posterior belongs to the same family with $\mathbf{m}_0$ and the v_i updated by addition — the “precisions add” structure of Eq. 7.6.2, the prior read as $\sum_i v_i$ observations at $\mathbf{m}_0$. What is missing is everything one wants afterwards: outside the normal family, and outside degenerate designs such as the intercept-only Poisson model of [Exercise \(B1\)](#), [Eq. 39.3.2](#) has no closed-form normalizing constant, so neither have the posterior mean, the variance, the marginal likelihood or a credible interval. In practice the normal prior of the next section — not conjugate, but interpretable — is what is used.

Idea. Everything in the Bayesian thread that was a *definition* survives; everything that was a *closed form* does not. The next two sections replace the closed form by an approximation accurate when n is large and a simulation accurate when it is run long enough.

39.3.4 Exercises

39.3.4.1 A. Check your understanding

Exercise A1.

Verify directly, from the form of the Poisson log-likelihood with a log link, that it is not a quadratic in β for $p = 1$, and identify the term that spoils it.

Solution. With $\mu_i = e^{x_i\beta}$ the log-likelihood is $\sum_i \{y_i x_i \beta - e^{x_i\beta}\}$ up to a constant. The first term is linear; the second has second derivative $-\sum_i x_i^2 e^{x_i\beta}$, which depends on β . That is the cumulant term $b(\theta) = e^\theta$, whose b'' is not constant, as [Proposition 39.3.1\(b\)](#) requires.

Exercise A2.

An equal-tailed credible interval for β_j transforms to an equal-tailed credible interval for e^{β_j} by exponentiating its endpoints. Is the same true of a highest-posterior-density interval? Explain.

39.3.4.2 B. Practice

Exercise B1.

Take the Poisson family with a log link, $p = 1$ and $x_{i1} = 1$ for all i , so the model has a single parameter β and $\mu = e^\beta$. Write out [Eq. 39.3.2](#) explicitly, show that it is the density of $\beta = \log \mu$ when μ has a gamma distribution, and identify the shape and rate in terms of $\mathbf{m}_0$ and the v_i . Why does the closed form appear here and not in general?

Solution. With $b(\theta) = e^\theta$, $\theta = \beta$ and $\kappa_0 = 1$, [Eq. 39.3.2](#) is $\pi(\beta) \propto \exp\{a\beta - ve^\beta\}$ with $a = \mathbf{1}^\top \mathbf{m}_0$ and $v = \sum_i v_i$. Substituting $\mu = e^\beta$, so $d\beta = d\mu/\mu$, the density of μ is proportional to $\mu^{a-1} e^{-v\mu}$: a gamma with shape a and rate v , normalizing constant $v^a/\Gamma(a)$. That constant is available because the single linear predictor lets the integral factor into one dimension. With $p > 1$ the exponent $\mathbf{m}_0^\top \mathbf{X}\beta - \sum_i v_i e^{\mathbf{x}_{(i)}^\top \beta}$ is a sum of exponentials of different linear forms, and no change of variable makes it a product of gamma integrals.

39.3.4.3 C. Going deeper

Exercise C1.

[Proposition 39.3.1](#) assumed the canonical link. Show that with a non-canonical link the conclusion still holds in the following sense: if the log-likelihood is a quadratic in β for every design and every response, then the working weights of Theorem 34.4.1 do not depend on β and the mean is affine in β . Deduce that the model is the normal linear model up to a reparameterization.